QR-001 Mix ratios and cure holds

Doc ID	QR-001
Title	Mix ratios and cure holds
Parent	CS-008 §5; CS-008 §5.1; CS-008 §6; CS-008 §7
Checked against	CS-008 Rev E; CS-008 Rev E; CS-008 Rev E; CS-008 Rev E

Derived from its parent standard. If this and the parent disagree, the parent is right.

1. IN2 resin and AT30 hardener mix ratio (AT30 SLOW or FAST)

IN2 resin and AT30 hardener mix 100:30 by weight, with AT30 SLOW or AT30 FAST (CS-008 §5). Weigh both on a scale to 1 g: IN2 publishes no ratio tolerance (CS-008 §5). AT30 SLOW is the infusion default.

Also called: IN2, AT30, hardener, slow hardener, fast hardener, AT30 ratio, infusion resin, mix ratio, how much hardener, 100 to 30.

Hardener	Ratio by weight	Pot life (datasheet)	Use
AT30 SLOW	100:30	120–150 min at 25 °C in the TDS table; the TDS text says 80–100 min, so plan on 80–100 min	Infusion default
AT30 FAST	100:30	10–14 min at 25 °C in the TDS table; the TDS text says 9–14 min, so plan on 9–14 min	Small infusions only

Source: the IN2 TDS, `03 Datasheets/EC-TDS-IN2-Infusion-Resin.pdf`, as copied into CS-008 §5.

1. Weigh resin and hardener on the 1 g scale (CS-008 §7 step 2).
2. Write both masses on the work order.
3. Mix for at least 2 min, scraping the walls and base (CS-008 §7 step 3).
4. Double-pot: pour into a second cup and remix with a fresh stick.

Want a pot life between SLOW and FAST? You may blend AT30 FAST and SLOW, as long as the total stays 100:30 by weight (CS-008 §6). Never mix off-ratio to speed up or slow down the cure. Off-ratio gives an uncured, sticky part.

2. West 105 mix ratio: how much 205 fast, 206 slow or 209 extra slow hardener (pumps and scale)

West 105 with 206 or 205 is 5:1 by volume on the pumps. West 105 with 209 is 3:1 by volume and needs a different pump head (CS-008 §5). Check the pump head before the first stroke: the wrong pump is the classic wrong-ratio trap (CS-008 §7 step 1).

Also called: West, West System, 105, 205, 206, 209, fast hardener, slow hardener, extra slow, pumps, pump ratio, mix ratio.

Hardener	Pumps (by volume)	By weight	Scale tolerance (by weight)	Pot life, 100 g at 72 °F
206 slow (wet layup default)	5:1	5.36:1	4.84–6.19:1	20–25 min
205 fast	5:1	5.19:1	4.83–6.20:1	9–12 min
209 extra slow	3:1	3.68:1	3.30–4.03:1	40–50 min

All figures are from the West TDSs, each row from its own 105/2xx sheet (CS-008 §5).

1. Check the pump head matches the hardener: 3:1 for 209, 5:1 for 205 and 206.
2. Pump full strokes only (CS-008 §7 step 2).
3. Write the strokes or masses on the work order.
4. Mix for at least 2 min, scraping the walls and base, then double-pot (CS-008 §7 step 3).

Do not blend 205 and 206: no datasheet on file covers it (CS-008 §6). After a long gap between uses, mix a small test batch to check the pumps meter correctly and the batch cures (CS-008 §7 step 9). Cloudy or crystallized 105 after a cold spell: warm it to 125 °F and stir until clear before it goes in the pump (CS-008 §7 step 8).

3. XCR mix ratio (mold sealing resin)

XCR mixes 100:41 resin to hardener by weight, which is 2:1 by volume (CS-008 §5). Pot life is 8 min at 20 °C for a 150 g mix, so mix small (XCR TDS, `03 Datasheets/EC-TDS-XCR-Epoxy-Coating-Resin.pdf`, as copied into CS-008 §5).

Also called: XCR, coating resin, sealer, mold sealer, seal coat, sealing resin.

- Weigh it to 1 g. XCR publishes no ratio tolerance (CS-008 §5).
- Minimum temperature: 15 °C (59 °F), the datasheet figure (CS-008 §5).
- FEB hold: 12 h before the release system goes on. That is the team's hold; the datasheet gives initial cure as 8 h 30, handleable at about 9 h at 20 °C (CS-008 §5).
- The app does not lock the XCR hold. No work-order step starts an XCR cure, so the 12 h is watched by hand (CS-008 §5.1).

Batch size, how many coats and the overcoat window are in QR-004.

4. How long before I can demould (demold)? Cure time and the FEB hold for each resin

Wait the FEB hold, counted from the end of the layup or infusion: 48 h for IN2 + AT30 SLOW, 24 h for IN2 + AT30 FAST, 24 h for West 105 with 206 or 205, 36 h for West 105 + 209, and 12 h for XCR before release (CS-008 §5).

Also called: demould, demold, de-mould, cure hold, FEB hold, wait time, cure time, how long to cure, when can I pull the part.

Resin (as the app names it)	FEB hold (team)	Datasheet figure (manufacturer)
IN2 + AT30 SLOW	48 h	demould 18–24 h at 25 °C
IN2 + AT30 FAST	24 h	demould 6–8 h at 25 °C
West 105 + 206 slow	24 h	thin-film solid 10–15 h at 72 °F
West 105 + 205 fast	24 h	thin-film solid 6–8 h at 72 °F
West 105 + 209 extra slow	36 h	thin-film solid 20–24 h at 72 °F
XCR coating resin	12 h	initial cure 8 h 30, handleable about 9 h at 20 °C

The FEB hold is the team's, signed off by Simon Starbuck 2026-08-01, and it is the one that governs. The datasheet column is the manufacturer's (CS-008 §5, §5.1). West publishes no demould time; its thin-film cure is the nearest figure.

On a work order, "Cure and demould" ("Cure" on a Foam wrapped run) stays locked until the hold has run from the "Finished on" and "at" time entered when "Infuse", "Wet layup and bag" or "Wet layup over core" was signed (CS-008 §5.1). To see the hold in force today, press "why N h?" on the held step. Gel is not cure: fingernail-test the flange before you trust the clock (CS-008 §7 step 6).

5. Why is the hold longer than the datasheet? ("Why N hours?")

The FEB hold is a team decision, longer than the datasheet in every row on purpose. It lands on shift boundaries: 24 h means "come back tomorrow" and 48 h means "come back the day after tomorrow", which is after the weekend only for a Friday infusion (CS-008 §5.1). It was never worked out from the datasheet.

Also called: why N hours, why 48 hours, why so long, FEB hold, team hold, can I demould sooner.

A hold that ran out at 3 am would help nobody, because nobody is at RFS at 3 am. That is why the ratios against the datasheet look uneven: 1.3× for XCR, 3× for the FAST systems (CS-008 §5.1).

Both FAST systems hold 24 h. Buying FAST hardener to turn two infusions round in one day does not work under this rule. That is deliberate: the team has no record of a part ruined by waiting, and several of work being rushed (CS-008 §5.1).

The holds were signed off by Simon Starbuck on 2026-08-01. The live value is the app's. "Why N hours?" always shows the hold in force and who signed it, and says when a lead set it from the app (CS-008 §5.1). QR-001 §6 says how a lead changes one.

6. Can a lead change the hold or demould (demold) early?

Yes. A lead can sign the locked step early, or change a hold for every work order, and the app records both. A member who tries to sign early is turned away, and no hold can go below the datasheet figure (CS-008 §5.1).

Also called: override, skip the hold, early demould, early demold, change the hold, lead override.

Demould one part early (lead only):

1. On the locked step, sign through "Demould early?".
2. Answer "Why are you demoulding early?".
3. The app writes the reason, the lead's name and how many hours short it was to the work order's event log.

Change a hold for every work order (lead only):

1. Open a work order whose hold for that resin is running now. On the held step, press "why N h?" to open "Why N hours?". The link shows only while a hold is counting down (CS-008 §5.1).
2. Press "Change this hold…".
3. Enter "FEB hold (hours)".
4. Under "Signed off by", pick who signed it off and the date.
5. Press "Save".

The app refuses a hold below the datasheet figure, and a hold nobody signed off. "Revert to the code table" in the same window undoes a change. A lead who changes a hold shall open a revision of CS-008 within two weeks, so the table and the app agree again (CS-008 §5.1).

The app does not lock three cases: XCR mold sealing (watched by hand, CS-004 §7.3), any cure recorded as "Something else / not recorded", and retro work orders (CS-008 §5.1).

7. Pot life and working time for each resin

IN2 + AT30 SLOW: plan on 80–100 min (the TDS text, no temperature given); AT30 FAST: plan on 9–14 min. West 105 + 206 20–25 min, + 205 9–12 min, + 209 40–50 min (100 g at 72 °F). XCR 8 min at 20 °C (150 g) (CS-008 §5).

Also called: pot life, working time, open time, gel time, how long do I have.

Pot life is the usable time in the cup at the mass and temperature the datasheet quotes it for. A bigger batch or a warmer shop makes it shorter (CS-008 §4). Pick a system whose pot life fits the job with a margin (CS-008 §7 step 1).

System	Pot life (datasheet)
IN2 + AT30 SLOW	plan on 80–100 min (the TDS text, no temperature given; its table says 120–150 min at 25 °C)
IN2 + AT30 FAST	plan on 9–14 min (the TDS text; its table says 10–14 min at 25 °C)
West 105 + 206	20–25 min at 72 °F (100 g)
West 105 + 205	9–12 min at 72 °F (100 g)
West 105 + 209	40–50 min at 72 °F (100 g)
XCR	8 min at 20 °C (150 g)

Sources: CS-008 §5, from the IN2, West and XCR TDSs in `03 Datasheets/`.

Mix the smallest batch that does the job and pour it out of the pot promptly: resin packed in a cup heats itself faster (CS-008 §6). A warm cup means use it now or spread it thin. A hot or smoking cup goes outside onto concrete and stays there. Do not bin it and do not pour water on it indoors. Nothing goes in the trash until it is fully cured and cool (CS-008 §6).

8. Shop too cold? Minimum temperatures and cold cures

Minimum cure temperatures (datasheet): West 206 60 °F (16 °C), 205 40 °F (4 °C), 209 70 °F (21 °C); XCR 15 °C (59 °F); IN2 + AT30 not stated. Cure at 20 °C or warmer where you can (CS-008 §5; §7 step 5).

Also called: cold shop, winter, temperature, minimum temperature, cure temperature, cold cure.

System	Minimum temperature (datasheet)
West 105 + 206	60 °F (16 °C)
West 105 + 205	40 °F (4 °C)
West 105 + 209	70 °F (21 °C)
XCR	15 °C (59 °F)
IN2 + AT30	not stated

Source: CS-008 §5, from the West and XCR TDSs.

The pot life and cure figures in CS-008 §5 assume 25 °C for IN2, 72 °F (22 °C) for West and 20 °C for XCR. A colder shop cures slower, and the hold does not stretch on its own, so test the flange, not the clock (CS-008 §5.1).

1. Cure at 20 °C or warmer wherever you can (CS-008 §7 step 5).
2. When you sign the layup or infusion, fill in “Shop temperature, °C (optional)”.
3. Below the reference temperature the app warns “Colder than this number assumes, so it will run long. Test the flange, not the clock.” The hold itself does not change (CS-008 §5.1).
4. Before you demould, fingernail-test the flange or trim zone for soft spots (CS-008 §8).

9. Mixing rules for every resin, and what to record

Weigh on the 1 g scale or pump full strokes, mix for at least 2 min scraping the walls and base, and double-pot anything going on a part surface (CS-008 §7). Never mix off-ratio to change the cure speed (CS-008 §6).

Also called: mixing, double pot, double-pot, scale, weigh, expiry, expired resin, Resin mixed, batch.

1. Check the expiry: the EXP date on the jug's label, or the Expires date on its lot page. Past it, ask a lead before it goes on a part (CS-008 §7 step 1).
2. Weigh to 1 g, or pump full strokes for West, and write the masses on the work order (CS-008 §7 step 2).
3. Mix for at least 2 min, scraping the walls and base (CS-008 §7 step 3).
4. Double-pot: transfer to a second cup and remix with a fresh stick.
5. When you sign the layup or infusion, pick the resin you actually mixed under "Resin mixed" (CS-008 §7 step 6).
6. Choose "Something else / not recorded" only for a resin not in the CS-008 §5 table. Then write the resin in the step note; its demould time is a lead's call.
7. Spread leftovers thin on scrap, or leave them outside to cure (CS-008 §7 step 7).

Forged CF mix math: the finished part mass is about cavity volume × 1.4 g/cc, the fiber is 60% of that (weigh it out first), and the resin to mix is about fiber mass × 1.25; the excess squeezes out (CS-008 §7 step 4). Starving the mix made the first forged part brittle; it failed its bend test.

10. When is the part fully cured? (full strength)

Full strength takes days to weeks, not hours: at least 14 days at 25 °C for IN2, at least 14 days at 20 °C for XCR, working strength in 1–4 days for West with 206 or 205, and maximum strength in 4–9 days for West with 209 (CS-008 §5).

Also called: full cure, fully cured, full properties, full strength, post-cure, when can I load it.

Demould is not full cure. A part out of the mold after its FEB hold is still curing. Avoid structural or fit loading before full cure where you can (CS-008 §4). The team cures ambient with no oven, so there is no post-cure to shorten this (CS-008 §5.1).